Bin Cai, Ph.D.

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Education

Ph.D., Iowa State University, Ames, IA, USA *2019*
Co-majors: Civil Engineering; Wind Energy Science, Engineering, and Policy
Dissertation: Enabling tall towers in wind energy production

M.S., Tongji University, Shanghai, China *2013*
Major: Civil Engineering (specialized in Structural Wind Engineering)
Thesis: Wind loads on low-rise buildings

B.S., Xiangtan University, Hunan, China *2010*
Major: Civil Engineering

Appointments

Department of Civil, Construction, and Environmental Engineering, Iowa State University
Postdoc Research Associate *2020 – present*
Graduate Teaching Assistant *2019 – 2020*
Graduate Research Assistant *2013 – 2019*

Siemens Corporate Core Division, Princeton, NJ
Graduate Research Intern *Feb. - May 2015*

School of Civil Engineering, Tongji University
Graduate Research Assistant *2010 – 2013*

Research Experiences

Research lead, material characterization for Ultra-High performance concrete (UHPC) and its structural applications, Iowa State University, Ames, IA *2023 – present*

- Develop numerical models for viable designs of tall towers using precast UHPC components and unbonded post-tensioning technique
- Develop viable designs of integral abutment using precast UHPC components and 3D printing technique for Accelerated Bridge Construction
- Conducted experimental analysis on the material properties of different UHPC overlay mixes designed for rehabilitation of bridge decks
- Conducted experimental analysis to investigate the mechanical properties of regular concrete with different recycled fibers
- Conducted experimental analysis on the optimal compositions to increase the self-sensing capability for UHPC
- Developed test and safety protocols for laboratory experiments on UHPC

Operation lead, technology transition team for a new wind energy technology, Iowa State University, Ames, IA *2020 – 2024*

- Assisted in writing an awarded project proposal to support the technology development for licensing and commercialization

- Established a viable commercialization plan for a new wind energy technology
- Collaborated with cross-functional teams to address technical or logistical challenges to ensure project's progress and deliverable comply with their schedule and budget

Research Assistant, Iowa State University, Ames, IA *2013 – 2019*

- Performed numerical and experimental analysis of fatigue performance for tower connections and system under cyclic loadings for a novel tower design taller than 100 m
- Evaluated the commercial feasibility of different tall tower technologies for the US and European markets by estimating and comparing the levelized cost of energy
- Assessed the potential wind energy production harvested from wind towers taller than 100 m

Graduate Research Intern, Siemens Corporate Core Division, Princeton, NJ *Feb. - May 2015*

- Programmed the design theories in MATLAB and optimized the design variables for a novel tower model to minimize the levelized cost of energy while ensuring structural suitability
- Evaluated the structural suitability requirements of the parametric tower design against economic benefit

Graduate Research Assistant, School of Civil Engineering, Tongji University *2010 – 2013*

- Assisted in developing a data acquisition system for field testing of wind pressure on a roof-adjustable test building
- Investigated probabilistic distribution models to characterize wind pressure and estimated the extreme values under monsoon and typhoon conditions

Selected Publications

1. B Cai, S Gali, P Kalmogo, and S Sritharan (2026). Evaluation of freeze-thaw damage to UHPC overlay mixes and their bonding to normal concrete. Engineering Structures (under review).
2. S Gali, B Cai, P Kalmogo, S Sritharan. (2026). Characteristics of UHPC overlay mixes suitable for bridge deck rehabilitation. ACI Materials. (under review).
3. B Cai, S Sritharan, P Gayes, and LJ Pietrafesa (2026). Tall onshore wind energy production in the US Southeast and cost effectiveness. ASCE Journal of Energy Engineering. (under review).
4. H Liu, S Laflamme, B Cai, P Lyu, S Sritharan, and K Wang (2024). Investigation of 3D printed self-sensing UHPC composites using graphite and hybrid carbon microfibers. Sensors, 24, 7638.
5. B Cai, P Vo, S Sritharan, ES Takle (2021). Wind energy potential at elevated hub heights in the US Midwest Region. ASCE Journal of Energy Engineering 147 (4), 04021023.
6. P Barutha, A Nahvi, B Cai, HD Jeong, S Sritharan (2019). Evaluating the commercial feasibility of a new tall wind tower design concept using a stochastic levelized cost of energy model. Journal of Cleaner Production 240, 118001.
7. S Sritharan, R Peggar, B Cai, S Lin, MC Hsu, J Krennrich, et al. (2016). Hexcrete tower for harvesting wind energy at taller hub heights – Budget period 2. Final report DOE-ISU—06737-1. Iowa State University, Ames, IA.
8. XF Yu, ZN Xie, X Wang, B Cai (2016). Interference effects between two high-rise buildings on wind-induced torsion. Journal of Wind Engineering and Industrial Aerodynamics 159, 123-133.

Conference Papers, Presentations, and Invited Talks

1. B Cai, S Gali, P Kalmogo, and S Sritharan (2026). Freeze-Thaw evaluation of UHPC overlay to rehabilitate concrete bridge decks. Third International Interactive Symposium on Ultra-High Performance Concrete. Des Moines, IA, USA. (Accepted).
2. B Cai, S Sritharan (2024). An evaluation of the interface shear capacity of concrete bridges using UHPC overlay. Iowa Department of Transportation UHPC technology exchange workshop. Ames, IA, USA. July 17-18, 2024.
3. B Cai, S Sritharan (2023). Assessment of fatigue load effects on UHPC components for wind energy applications. Third International Interactive Symposium on Ultra-High Performance Concrete. Wilmington, DE, USA. June 4-7, 2023.
4. S Sritharan, B Cai, and P Gayes (2023). Feasibility of land-based wind generated electricity for the US Southeast. 2023 Global Energy Meet (GEM). Boston, Massachusetts.
5. B Cai, S Sritharan, P Gayes, and LJ Pietrafesa (2023). Land-Based wind power generation potential for South Carolina. 2023 SC EPSCoR State Conference. Summerville, SC, USA. April 14, 2023.
6. B Cai, S Sritharan (2023). Assessment of tall onshore wind benefits when integrating with other renewables in the US Southeast. American CLEANPOWER Conference & Exhibition, New Orleans, LA, USA. May 23-24, 2023.
7. B Cai, S Sritharan (2022). Onshore wind energy potential in Sri Lanka. International Conference on Engineering, University of Jaffna, Jaffna, Sri Lanka. August 19, 2022.
8. B Cai, S Sritharan (2022). Cost/Benefits comparison of offshore wind and tall onshore wind in the US Southeast. American CLEANPOWER Conference & Exhibition, San Antonio, TX. May 16-18, 2022.
9. B Cai, P Vo, S Sritharan, ES Takle (2019). An evaluation of the energy production in Iowa at elevated heights. Electric Power Research Center (EPRI) Symposium. Ames, IA, USA, May 19, 2019.
10. B Cai, ES Takle, S Sritharan (2018). Opportunities to broaden deployment of tall wind towers in the US wind-rich regions. Structural Engineering Seminar. Iowa State University. Ames, IA, USA, February 28, 2018.
11. ES Takle, B Cai, S Sritharan (2017). Wind energy potential at six locations in the Midwest US at higher hub heights. International Conference on Future Technologies for Wind Energy. Boulder, CO, USA, October 24-26, 2017.
12. B Cai, P Vo, S Sritharan, ES Takle (2017). An evaluation of wind energy production in Iowa at elevated heights. North American Wind Energy Academy (NAWEA) Symposium. Ames, IA, USA, September 26-28, 2017.
13. S Musuvathy, A Parlak, S Srivastava, L Mirabella, V Brandstetter, EB Arisoy, B Cai, TL Tang (2016). Design optimization for a wind turbine tower. Tall tower commercialization workshop. Denver, CO, USA, June 23, 2016.
14. B Cai, H Wibowo, S Sritharan (2016). Experimental fatigue performance evaluation of Hexcrete tower connections. Iowa Energy Center. Ames, IA, USA.
15. B Cai, S Sritharan (2015). Experimental fatigue performance evaluation of Hexcrete tower connections. Wind Energy Industry Symposium. Ames, IA, USA, September 28, 2015.

Funding and Awards

1. Title: Enabling Hexcrete Technology for Tall Wind Towers Nationwide

Role: Proposal lead contributor
PI: Dr. Sri Sritharan (advisor)
Sponsor: Iowa Regents Innovation Fund
Amount: $50,000
Duration: 06/2022 – 12/2023

2. Title: Market discovery for Hexcrete technology in the renewable energy sector

Role: Proposal lead contributor and entrepreneurial lead
PI: Dr. Sri Sritharan (advisor)
Sponsor: ISU I-Corps Minigrant Funds
Amount: $3,000
Duration: 11/2021 – 10/2022

3. Title: Commercialization plan for Hexcrete Renewables. (2022 Semifinalist)

Role: Proposal lead contributor and entrepreneurial lead
PI: Dr. Sri Sritharan (advisor)
Sponsor: John Pappajohn Iowa Entrepreneurial Venture Competition

4. ISU Active Learning Institute professional development grant *Fall 2022*
5. Earthquake Engineering Research Institute Student Leadership Council travel grant *April 2016*

Professional Development

1. Spring 2026 Postdoc Researcher Professional Development Grant
2. Fall 2025 cohort of Preparing Future Faculty Workshop from Auburn University
3. Summer 2023 and Summer 2024 cohort of Inclusive Innovators Summit
4. Summer 2023 cohort of CELT Inclusive Course Design Institute
5. Team lead in the 2023 ISU Go-To-Market program
6. Fall 2022 cohort of CELT Active Learning Institute
7. Team lead in the Spring 2022 ISU Startup Factory program
8. Team lead in the Fall 2021 ISU Innovation Corps program

Professional Society Memberships

1. American Concrete Institute (2014 – 2020, 2025 – present)
2. American Clean Power Association Clean Energy Demand Committee (2023 – present)
3. Iowa Association for Energy Efficiency (2023 – 2024)
4. American Society Civil Engineering (2014 – 2020)
5. Earthquake Engineering Research Institute (2015 – 2018)

Technical Reviews

1. Concrete structure related: Transportation Research Board Annual Meeting (2025 – present); Construction and Building Materials (2024 – present); International Interactive Symposium on Ultra-High Performance Concrete. (2023, 2026)
2. Energy related: Applied Energy (2025 – present); National Conference on Undergraduate Research (NCUR), Council on Undergraduate Research (2023 – present); Energy Systems (2023 – present)

Teaching and Mentoring Experiences

Teaching Assistant, Iowa State University, Ames, IA

1. Graduate course of Structural Dynamics (7 students). Spring 2025.
2. Graduate course of Prestressed Concrete Design (7 students). Fall 2021.
3. Graduate course of Wind Energy Systems (5 students). Fall 2021.
4. Undergraduate course of Reinforced Concrete Design (94 students). Spring 2019.

Undergraduate and graduate research mentee, Iowa State University, Ames, IA

1. Asish Chaulagain, graduate researcher. Fall 2025 - present.
   Topic: Durable integral abutment using stay-in-place UHPC shells for Accelerated Bridge Construction.
2. Jozepha Soborowicz, undergraduate researcher. Summer and Fall 2024.
   Topic: UHPC overlay mixing and the bond strength between UHPC overlay and concrete substrate.
   Outcome: a) Completion of a set of UHPC mixing experiments and UHPC overlay pull-off tests, and b) Bond strength of UHPC overlay for bridge decks, oral presentation at UHPC Workshop with the US Department of Transportation from multiple states in the Midwest.
3. Annalucia Mendoza Lezama, undergraduate research intern. Summer 2024.
   Topic: Numerical modeling of tall wind turbine towers.
   Outcome: Completion of a set of design drawings for tall wind turbine towers.
4. Rahul Bhat, undergraduate researcher. Fall 2023.
   Topic: Evaluation of material properties of Ultra-High Performance Concrete.
   Outcome: Completion of a set of UHPC direct tensile tests.
5. Jian Janes, Hunter Borys, Emily Jennings. Undergraduate researchers in the ISU CyBIZ Program. Spring 2023.
   Topic: Evaluating the benefits of tall wind towers in the US and Hexcrete market research.
   Outcome: Submission of a Hexcrete technology commercialization plan to the US Department of Energy 2023 EnergyTech University Prize competition.
6. Ryan Casey, undergraduate researcher in the First Year Honors Mentor (FYHM) Program. Spring 2022.
   Topic: Evaluating the benefits of tall wind towers in the US.
   Outcome: FYHM mentor grant.
7. Claire Schneider, undergraduate researcher in the First Year Honors Mentor (FYHM) Program. Spring 2022.
   Topic: Wind energy & the impact of taller towers.
   Outcome: FYHM mentor grant.
8. Zexi Yin, undergraduate researcher. Fall 2021 and Spring 2022.
   Topic: Wind energy and evaluation of wind load on tall towers.
   Outcome: a) Evaluation of wind loads on Hexcrete wind turbine towers, oral presentation at the National Conference on Undergraduate Research, and b) The current development and future trends of wind energy, poster presentation at the ISU 16th Symposium on Undergraduate Research and Creative Expression.
9. Phuong Vo, undergraduate researcher in the NSF Research Experiences for Undergraduates (REU). Summer 2019.
   Topic: Wind energy production for tall towers.
   Outcome: Benefits of tall wind towers, oral presentation at the REU Summit.

Educational Services and Outreach

1. Graduate advisor, American Concrete Institute ISU student chapter. (2025)
2. Graduate advisor, the ISU Collegiate Wind Competition team (Payton Bills, Drew Jensen, Jozepha Soborowicz). The US Department of Energy 2025 Collegiate Wind Competition. Boulder, CO. (2025)
3. Judge, 3MT thesis presentation, Iowa State University. (2024-2025)
4. Graduate advisor, the ISU team (Jian Janes, Hunter Borys) for the US Department of Energy Office of Technology Transitions 2023 EnergyTech University Prize (2023).
5. Graduate advisor, the ISU Collegiate Wind Competition team (Isaac Twedt-Ball, Payton Bills, Drew Jensen, Simon Hoefing, Noah Gerken, Abigayle Moser). The US Department of Energy 2023 Collegiate Wind Competition. Boulder, CO. (2023)
6. Innovator Summit Coordinator, ASCE Innovator Summit at ASCE Mid-America Student Symposium. Ames, IA. (2022).
7. Graduate advisor, ISU Wind Energy Student Organization (2017)
8. Competition Organization Committee, Undergraduate Seismic Design Competition. San Francisco, CA. (2016)
9. Student Leadership Council, Earthquake Engineering Research Institute (2015 – 2016)
10. President, ISU student chapter of Earthquake Engineering Research Institute (2015 – 2016)
11. Table lead, Elementary Engineering program hosted by Story County 4-H & Youth in Iowa (2015)
12. Guest speaker, Science and Technology Night for Fellows Elementary School (2015)

Skills

- Subject expertise: reinforced concrete, structural dynamics, nonlinear analysis, experiment design, finite element modeling, renewable energy potential, and techno-economic analysis
- Engineering software and programming language: ANSYS, SAP2000, Abaqus, LPile, JMP, AutoCAD, SolidWorks, NREL SAM, Furow, Matlab, Python
- Structural engineering codes or standards: ACI 318, AISC, AASHTO LRFD, IEC61400-1, CEB-FIP Mode Code 2010, ASCE 7